

January 26, 2026 06:00 AM GMT

Quantitative Investment Strategies

Quantitative Strategies in 2026:
Continuity, Convexity and
Opportunity

We provide an outlook for Quantitative Investment Strategies (QIS) in 2026. We recap performance in 2025 and use this information as a guide for 2026. Further fundamental aspects lead to a generally constructive QIS return outlook.

QUANTWISE 

QIS performance in 2025 was volatile but ultimately constructive, laying a positive foundation for 2026: Despite some drawdowns around 'Liberation Day', the majority of QIS strategies ended the year in positive territory, with equities, commodities and volatility contributing strongly. Defensive overlays generally worked as intended during the tariff-driven sell-off, while faster-reacting convex strategies outperformed trend following during abrupt moves. Overall resilience in a difficult macro environment provides confidence as markets transition into 2026.

The 2026 macro environment mirrors late 2025 conditions, creating a broadly supportive backdrop for systematic strategies: Growth remains supported by fiscal policy, easing financial conditions and AI-driven investment. Moderating inflation enables earlier rate cuts in the US and further easing in Europe, while equity markets enter a 'risk reboot' phase marked by stronger breadth and earnings-driven returns. Rates markets should exhibit front-end declines and mild re-steepening.

Systematic forecasting frameworks point to generally positive QIS expected returns, with equity factors, volatility carry, realized skewness and commodity curve carry leading the outlook: The ATLAS framework shows predominantly positive forward-return signals, especially in equity multi-factor and quality, volatility carry across markets, commodity curve carry and cross-asset realized skewness. Economic similarity analysis reinforces the expectation of positive performance, particularly across equities, commodities and volatility. The combined message: 2026 likely extends the favorable dynamics observed in late 2025.

Trend following, equity systematic strategies and volatility carry each have fundamental reasons for optimism: Trend following should benefit from falling front-end rates and yield curve steepening, reversing prior headwinds; combining trend with realized skewness further enhances risk-adjusted returns. Equity factors remain appealing given high stock-specific risk, low correlations and AI-linked exposure within visionary quality. Volatility carry is supported by an attractive volatility risk premium, generally at or above historical averages. Investors may consider adding convex strategies to buffer return spikes due to geopolitical uncertainty.

MORGAN STANLEY & CO. INTERNATIONAL PLC+

Stephan M Kessler

Quantitative Analyst

Stephan.Kessler@morganstanley.com

+44 20 7425-2854

MORGAN STANLEY & CO. LLC

Vishwanath Tirupattur

Strategist

Vishwanath.Tirupattur@morganstanley.com

+1 212 761-1043

Aristeidis Tentes

Quantitative Analyst

Aris.Tentes@morganstanley.com

+1 212 761-1235

Krystal Qian

Quantitative Analyst

Krystal.Qian@morganstanley.com

+1 212 761-1788

MORGAN STANLEY & CO. INTERNATIONAL PLC+

Mingyong Liu

Quantitative Analyst

Mingyong.Liu@morganstanley.com

+44 20 7677-0963

MORGAN STANLEY EUROPE S.E., COPENHAGEN BRANCH+

Jakob Christian Lorenzen

Quantitative Analyst

Jakob.Lorenzen@morganstanley.com

+45 32 75 57-14

The performance data provided is a hypothetical illustration of mathematical principles; it does not predict or project the performance of an investment or investment strategy. Past performance is no guarantee of future results. QuantWise highlights research that incorporates a robust quantitative approach in our investment analysis.

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Executive Summary

In this section we summarize the key components of our outlook. We come up with views on QIS performance in 2026 from three different angles:

- Finding periods with economic similarities to the expected environment of 2026 and mapping return expectations to realized returns in those periods;
- ATLAS framework for the tilting of QIS;
- Additional asset class-specific considerations.

These frameworks complement each other by looking at different return horizons. ATLAS is a medium-term framework with efficacy of predominantly one but up to three-month forecasting horizons. The analysis of economic similarities has a one-year forecasting horizon.

Based on these analyses, we arrive at a set of QIS that stands out:

Exhibit 1: Outstanding QIS

Strategy	Ticker	Summary
Equity Multi Factor Portfolio	MSQEAFGU	Large amount of idiosyncratic volatility in single name equity markets, providing ample opportunities for alpha generation. Resilient returns in market drawdowns. Strong overall outlook by ATLAS framework, strong performance in similar economic periods.
Equity Visionary Quality	MSQEAQGU	Large amount of idiosyncratic volatility in single name equity markets, providing ample opportunities for alpha generation. Resilient returns in market drawdowns. Ability to participate in pro-cyclical markets through the "visionary" signals. Overlap with AI theme. Strong overall outlook by ATLAS framework, strong performance in similar economic periods.
Enhanced Trend Following	MSCBCTET	Headwinds to trend following about to abate with rates market turning more supportive for trend strategies. Market environment in 2026 expected to be similar to 2H 2025 when trend following performed well.
Enhanced Trend Following + Skewness	MSCBCTET + MSSKEWXA	Skewness is a strategy with strong outlook when using ATLAS framework and looking at similar economic periods. Strong performance in 2025 which we expect to carry over into 2026 due to a similar environment. Thus, it is a great exposure to hold in its own right. Diversifies trend following when the strategy becomes overstretched.
Commodity Curve Carry	MSCB2EXB	Strong return indication in ATLAS and economically similar periods. Strong 2025 return.
Broad Vol Carry Portfolio	MSCBVCB4	Implied volatility levels, slope, and volatility risk premium generally around long term averages or better. Overall environment leading to return expectations around historical averages, making short volatility in equities, fixed income, and commodities attractive. ATLAS framework, volatility timing framework, and economic similarity are all supportive.
Broad Vol Carry Portfolio + Long Convexity	MSCBXAWF	Based on our positive outlook for diversified volatility carry portfolios, we suggest the addition of long convexity. The case for added protection is driven partially by the bear case risk outlined in our economic and strategy outlook. Strategies protecting against shallow gaps down appear to be attractively priced when comparing volatility markets with the fundamental outlook. We particularly like long gamma, conditional down variance, and dispersion.
Brent Volatility	MSVCM1CO	Positive overall ATLAS score. Strong return dynamics in 2H 2026.
Enhanced Gold Beta	Custom	Long enhanced gold beta using short straddles and long call spreads. Benefits from gold prices with risk skewed higher (compare note Gold: Bull Case in Play, January 23, 2026). Provides defensive returns in terms of market distress with gold price up-moves.

Source: Bloomberg L.P., QIS Research

While we have conviction in our outlook, we want to highlight some limitations. Coming up with a return outlook on any market is a challenging endeavor. For QIS this is particularly hard as one needs to combine information on strategies with forecasts on the markets a strategy operates in. This leads to exposure to two sources of prediction error – we can be wrong about the underlying markets as well as the strategy behavior. That said, we believe there is value in tilting portfolios to strategies with a higher return expectation as long as one does so carefully. Our work shows that the tilting of QIS portfolios can add up to 0.2 in Sharpe ratio. This is a noticeable improvement and the context in which our outlook should be seen.

We will dive deeper into the specific drivers in the upcoming sections.

Performance in 2025 Volatile but Overall Positive

In light of the volatile nature of financial markets in 2025, it comes as no surprise that we also find a wide variation in returns across QIS. Overall, more than half of our baseline strategies generated positive returns in 2025 ([Exhibit 2](#)). The median Sharpe ratio is a respectable 0.26. When we look at the 2025 returns until April 8, 2025 we find a less constructive picture. Only 24% of the strategies are generating positive returns. From April 8, 2025 onwards until the end of the year we find that this number increases to 68%, illustrating a rather strong performance of QIS during the last eight months of 2025.

Positive average returns are achieved across commodities, equities, volatility and cross-asset strategies. Negative returns are observed for the whole year on average in FX and rates strategies, no doubt driven by an environment that remained volatile past the peak of market uncertainty in April 2025.

Median Sharpe ratios across strategies are strong in equity and volatility strategies, with values of 0.75 and 0.60, respectively. Commodity QIS delivered an outstanding performance with a median return Sharpe ratio of 1.36. Cross-asset skewness achieved an impressive realized Sharpe ratio of 3.33.

Focusing on styles, value is the clear laggard with a mean return of -4.1% across our strategy selection. For momentum and carry, we obtain values of 2.5% and 2.4%, respectively. For momentum this result is predominantly driven by outsized returns we find in equities and commodities momentum (mainly during April-December 2025). On the carry side, volatility carry does particularly well on average across markets. Commodity curve carry also delivers strong 2025 return at 10.9%.

Thus, **performance of QIS was generally encouraging in the difficult market environment of 2025**. In the next sections we will spend some time understanding what this may imply for 2026.

While a broadly diversified portfolio of QIS with a focus on return generation fared positively in 2025, we need to look at the performance of convex QIS separately. Over recent years, the use of QIS for diversification and protection in broader portfolios has become popular among investors. Thus, Liberation Day in April 2025 and the run-up to those events provide a rather interesting case study if those strategies held what they promised.

Exhibit 2: Performance overview of QIS in 2025

	Tickers	Return					Volatility	Sharpe Ratio
Start Date		31-Dec-24	18-Feb-25	02-Apr-25	08-Apr-25	31-Dec-24	31-Dec-24	30-Dec-24
End Date		18-Feb-25	08-Apr-25	08-Apr-25	31-Dec-25	31-Dec-25	31-Dec-25	31-Dec-25
Comdty: Carry (Cross Markets)	MSCBCCAG	-0.8%	-4.3%	-1.0%	9.1%	3.6%	9.5%	0.38
Comdty: Curve Carry	MSCB2EXB	-1.3%	3.8%	0.2%	8.3%	10.9%	7.5%	1.45
Comdty: Momentum	MSCBCMOG	0.8%	-1.1%	-1.3%	11.7%	11.3%	7.9%	1.43
Comdty: Pair Value	MSCBVB30	0.5%	-3.1%	-1.8%	-0.9%	-3.4%	8.7%	-0.39
Comdty: Realized Skewness	MSSKEWC0	-2.9%	4.8%	2.5%	15.1%	17.1%	6.5%	2.61
Comdty: Trend	MSCBTHDC	3.6%	-1.7%	-0.7%	9.9%	11.9%	8.8%	1.36
Comdty: Value	MSCBCVAG	-5.9%	0.2%	1.3%	-0.9%	-6.6%	9.0%	-0.73
Equity: Low Risk	MSQEALGU	1.3%	6.4%	-2.3%	-10.4%	-3.4%	12.7%	-0.27
Equity: Momentum	MSQEAMGU	1.8%	-0.4%	-1.2%	8.7%	10.1%	10.8%	0.94
Equity: Multi Factor	MSQEAFGU	0.5%	1.5%	-2.6%	6.1%	8.2%	8.2%	1.00
Equity: Quality	MSQEAQGU	0.0%	-0.7%	0.8%	6.4%	5.7%	10.1%	0.56
Equity: Realized Skewness	MSSKEWE0	0.5%	-3.8%	-1.9%	29.6%	25.2%	8.3%	3.03
Equity: Value	MSQEAVGU	-2.1%	-1.3%	-3.1%	6.2%	2.6%	10.0%	0.26
FX: Carry DM	MSCFCAD1	-0.8%	-7.0%	-7.3%	8.8%	0.4%	10.0%	0.04
FX: Carry EM	MSCFCAE1	-0.3%	-0.1%	-0.1%	-1.8%	-2.2%	6.0%	-0.36
FX: DM Reversion	MSCFMRD1	2.2%	-5.8%	0.6%	2.1%	-1.7%	8.1%	-0.21
FX: Equity-driven momentum DM	MSCFMED1	-1.9%	8.3%	-0.6%	8.8%	15.6%	11.2%	1.39
FX: Equity-driven momentum EM	MSCFMEE1	-2.5%	-5.5%	-3.6%	4.1%	-4.0%	10.6%	-0.38
FX: Realized Skewness (DM)	MSSKEWFD	-3.0%	-3.1%	0.2%	-7.0%	-12.6%	13.8%	-0.92
FX: Realized Skewness (EM)	MSSKEWFE	-0.2%	-0.1%	0.0%	3.9%	3.5%	6.9%	0.51
FX: Trend Global	MSCBCTAF	-3.9%	-3.9%	-1.2%	5.4%	-2.7%	10.4%	-0.26
FX: Value DM	MSCFVAD1	2.5%	0.2%	1.7%	-13.3%	-11.0%	8.7%	-1.26
FX: Value EM	MSCFVAE1	-1.7%	-0.2%	-0.4%	-8.3%	-10.0%	8.2%	-1.22
Rates: Carry (Cross Markets)	MSCBRACAG	-3.0%	2.3%	3.9%	-7.2%	-8.0%	10.5%	-0.76
Rates: Momentum	MSCBRMOG	-2.5%	-3.5%	-3.0%	-0.6%	-6.5%	8.6%	-0.75
Rates: Realized Skewness	MSSKEWB0	5.8%	-2.8%	-3.0%	10.9%	14.1%	9.9%	1.42
Rates: Short Rates Trend	MSQTSTRE	-0.4%	-2.7%	-0.2%	-3.2%	-6.2%	6.9%	-0.90
Rates: Trend	MSCBRTRG	0.4%	-2.9%	-3.0%	-3.6%	-6.0%	7.8%	-0.77
Rates: Value	MSCBRVAG	0.8%	-4.4%	-1.1%	7.5%	3.6%	9.3%	0.39
Vol: Brent	MSVCM1CO	1.4%	-2.7%	-3.1%	7.5%	6.0%	6.6%	0.92
Vol: Credit HY	MSCBVMHH	-0.3%	-5.8%	-6.2%	12.0%	5.2%	7.6%	0.69
Vol: SPX Vol Carry (1wk)	MSCBWGSP	0.4%	-8.9%	-9.4%	9.0%	-0.3%	8.9%	-0.04
Vol: SPX Vol Carry (3m)	MSCBVMSP	-0.4%	-6.0%	-3.7%	12.2%	5.0%	8.4%	0.60
Vol: SPX Vol Carry (Upside Vol)	MSCBUVS1	1.8%	-7.1%	-4.6%	-1.6%	-7.0%	12.3%	-0.57
Vol: TV Vol Carry	MSFVWGTT	-2.6%	-10.1%	-7.3%	17.1%	2.4%	9.7%	0.25
Vol: VIX	MSVXCS02	2.8%	0.7%	-3.4%	9.4%	13.2%	8.6%	1.53
Cross Asset: Skewness	MSSKEWXA	1.0%	-2.0%	-1.5%	25.9%	24.6%	7.4%	3.33
Cross Asset: Trend	MSCBCTET	2.9%	-9.5%	-5.2%	6.2%	-1.0%	9.3%	-0.11
% Positive		47.4%	23.7%	23.7%	68.4%	55.3%	NA	55.3%
Median								0.26

Source: QIS Research; Note: All strategies are scaled to an annualized volatility of 10% using long-term realized volatility to ensure comparability. Data as of December 31, 2025.

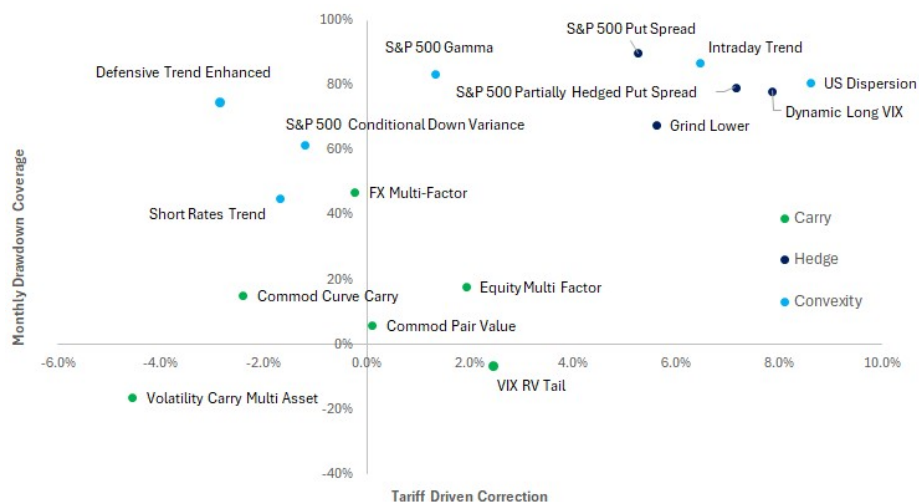
In [Exhibit 3](#), we show the long-term monthly drawdown coverage¹ as well as the performance of strategies during the tariff-driven market volatility around February-April 2025.

Strategies that are defensive in nature are highlighted in bright or dark blue. We see that most strategies such as intraday trend or dynamic long VIX are in the first quadrant,

1. A metric we use to understand the defensiveness of a strategy. It is the negative of the average return of a strategy in periods where the market is down by more than one standard deviation divided by the average market return in those periods. We volatility scale market and strategy returns to the same volatility level.

indicating that they provide defensiveness in the long term but also showed positive returns in the tariff-driven correction of February-April 2025. Thus, while the sell-off in 2025 was particularly dramatic – at least towards the end – we still find strategies to deliver defensiveness.

Exhibit 3: Comparing performance of defensive QIS in the long term versus the 2025 market sell-off



Source: QIS Research; Note: Strategies scaled to 6% annualized volatility. Data as of December 31, 2025.

In light blue we have three strategies which delivered less defensiveness that time. However, the nature of the sell-off provides the context to rationalize those returns: trend following is a strategy type that needs a sell-off to occur over a more prolonged period. Thus, trend following with its slower-adjusting positioning was not well suited for the abrupt sell-off we observed in April 2025 – and we would also not expect it to be. Conditional down variance is a strategy that becomes reactive only when markets drop deeply. Until April 8, the markets had not reacted enough for conditional down variance to show much defensiveness. However, after April 8 the positioning of the strategy provided a significant amount of positive convexity that offered strong returns when markets rebounded on April 9, but would have also delivered a strong performance if markets would have gone down further. Thus, conditional down variance also performed within our expectations.

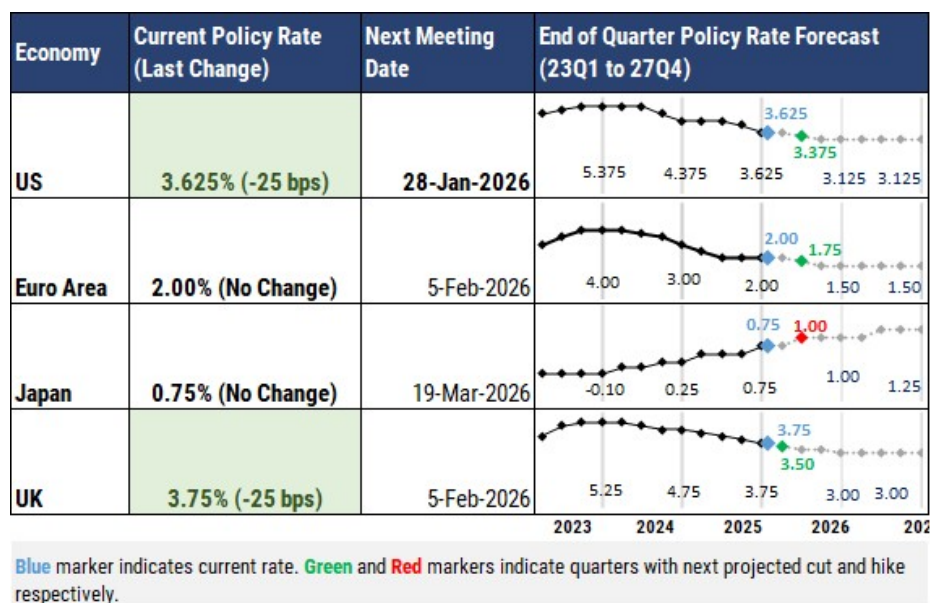
In summary, it is rather encouraging to see that most defensive strategies delivered positive returns in the sell-off around Liberation Day. A defensive overlay portfolio would therefore have done well. The defensive strategies that did not perform positively (e.g., trend, conditional down variance) delivered returns in line with expectations in this specific drawdown. **This result shows that the concept of the individual defensive strategies was indeed successful, but a deep understanding of how they work and interact with each other is necessary to assemble effective defensive overlays.**

Economic Backdrop

Before diving into specific QIS and their outlook, in this section we spend some time summarizing the expected economic and market environment. This environment is relevant as it sets the backdrop for the markets in which QIS are operating.

The global economy enters 2026 at a critical juncture. In some ways, the forecast contours of 2026 look similar to 2025, notably the pattern of sequential growth improving across the year. A key difference is that at the outset of the new year, global central banks – the Fed, the ECB and the BoJ – are at or near 'neutral', which presents its own uncertainty. The closer they get, the harder it is to find agreement on whether and in which direction policy rates should move (see [Exhibit 4](#) for our policy rates outlook). As Seth Carpenter, Morgan Stanley chief global economist notes, finding an equilibrium will not be easy and we expect markets to question the viability of any equilibrium once it is reached. Our macro strategists' views on interest rates also frame 2026 as a transitional year for global rates – from synchronous tightening to asynchronous normalization as central banks approach equilibrium, resulting in their call for DM government bond yields to remain range-bound.

Exhibit 4: Policy rate outlook

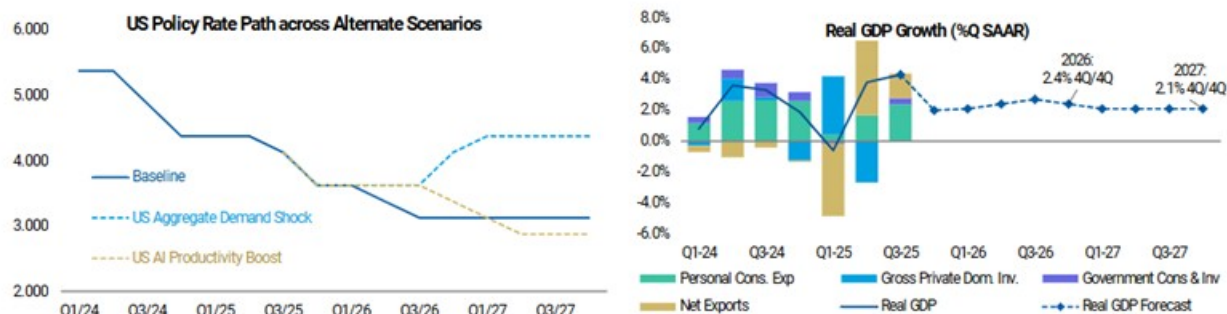


Source: Bloomberg, Morgan Stanley Research forecasts; Note: Data as of January 23, 2026.

That said, in the US, greater convergence of growth-supportive fiscal, monetary and regulatory policies combined with the tailwinds of productivity improvements and the AI-related capex bode positively for economic growth in 2026. Our economists have revised up their forecast and now pencil in a robust 2.4% (4Q/4Q) GDP growth for 2026.

Globally growth remains stable around trend, Europe struggles with low nominal momentum and an inflation undershoot, Japan experiences a reflationary rebound and emerging markets see gradual moderation rather than deterioration.

Exhibit 5: US GDP outlook



Against this macro backdrop, the outlook for equities is constructive and seen in a transition into a 'risk reboot' phase (Exhibit 6). The alignment of easing fiscal, monetary and regulatory policy creates a pro-cyclical environment in which asset class-specific fundamentals matter more than broad macro fears. US equities remain the preferred exposure, supported by strong operating leverage, above-consensus earnings growth and AI-driven efficiency gains. Importantly, we expect leadership to broaden beyond a narrow group of mega-cap stocks, improving market breadth and factor dispersion. Europe offers more selective opportunities amid rising dispersion, while emerging markets are viewed favorably in regions with strong domestic demand and financial sector leverage to growth. Overall, we expect equity returns to be driven increasingly by earnings and micro fundamentals rather than valuation expansion. However, our bull and bear case scenarios illustrate the substantial uncertainty around this outlook by having a particularly wide range (from 5,600 to 9,000 for the S&P 500). We observe similarly large ranges in MSCI Europe, Topix and MSCI EM, illustrating that our equity strategists have an overall constructive equity market outlook but see substantial risks associated with this outlook.

In rates markets, the key theme is a regime shift from aggressive easing towards neutrality. We expect US Treasury yields to decline further in 1H26 as Fed cuts are delivered, with the 10-year yield falling towards the mid-3% range, before rebounding later in the year as growth stabilizes and real rates rise for fundamentally positive reasons. Similar dynamics are expected across other developed markets, though with important divergences: we expect the ECB to ease further in response to persistent inflation undershoots, while the BoJ is expected to hike once more in 2026, allowing Japanese yields to stabilize at higher levels. This path implies meaningful curve and timing considerations for fixed income investors, with duration attractive early in the year but less so as growth improves. Credit markets reflect a more pro-cyclical environment, with Morgan Stanley favoring high yield over investment grade, expecting manageable default risk alongside increased capex and M&A activity.

Exhibit 6: Morgan Stanley Research forecasts for 4Q26

	As of Jan 16, 2026	Q4 2026 Forecast			Q4 2026 Return Forecast			Volatility	Base Case Return/Risk
		Bear	Base	Bull	Bear	Base	Bull		
Equities									
S&P 500	6,940	5,600	7,800	9,000	-18.2%	13.5%	30.8%	18%	0.74
MSCI Europe	2,446	1,920	2,430	2,700	-18.6%	2.2%	13.2%	15%	0.15
Topix	3,659	2,440	3,600	4,250	-31.3%	0.4%	18.2%	19%	0.02
MSCI EM	1,485	960	1,400	1,800	-33.1%	-3.5%	23.4%	16%	-0.22
FX									
JPY*	158	163	151	125	-6.3%	1.6%	23.7%	9%	0.17
EUR	1.16	1.05	1.16	1.36	-11.0%	-1.5%	15.7%	7%	-0.21
GBP	1.34	1.21	1.29	1.45	-9.5%	-3.5%	8.4%	9%	-0.38
AUD	0.67	0.61	0.67	0.75	-8.3%	0.7%	12.6%	10%	0.07
INR*	90.9	92.0	88.4	86.0	1.9%	5.5%	8.2%	5%	1.18
ZAR*	16.4	18.7	17.8	16.4	-9.5%	-5.1%	2.8%	15%	-0.33
BRL*	5.37	6.20	5.20	4.75	-4.8%	11.8%	21.6%	15%	0.76
MXN*	17.6	21.0	18.2	17.5	-12.3%	0.9%	4.4%	13%	0.07
Rates (% percent)									
UST 10yr	4.22	5.15	4.05	2.65	-2.2%	6.5%	18.1%	7%	0.97
DBR 10yr	2.83	2.90	2.45	2.05	2.8%	6.6%	10.1%	6%	1.17
UKT 10yr	4.40	4.65	3.90	3.25	3.1%	8.9%	14.2%	7%	1.36
JGB 10yr	2.18	2.30	1.65	1.05	1.8%	7.4%	12.8%	3%	2.92
Credit (bps)									
US IG	74	140	95	70	-3.8%	-0.7%	1.0%	3%	-0.23
US HY	251	450	300	250	-4.1%	0.4%	2.1%	6%	0.08
EUR IG	75	150	90	60	-2.6%	0.1%	1.4%	2%	0.04
EUR HY	251	450	280	225	-4.9%	0.7%	2.5%	5%	0.15
EM \$ Sov	248	400	260	225	-7.8%	1.5%	3.9%	7%	0.22
Sec. Products (bps)									
US Agency MBS	16	50	25	15	-1.9%	-0.4%	0.2%	2%	-0.19
US CMBS AAA	68	110	75	60	-3.5%	0.0%	1.4%	2%	-0.03
US CMBS BBB-	425	750	475	390	-28.1%	-0.9%	7.5%	4%	-0.25
Commodities									
Brent	64	50	60	120	-19.5%	-3.4%	93.2%	38%	-0.09
Copper	12,865	9,400	11,500	13,800	-26.3%	-9.9%	8.2%	21%	-0.47
Gold	4,611	4,080	4,800	5,760	-14.3%	0.9%	21.0%	15%	0.06

Source: Bloomberg, MSCI, Morgan Stanley Research forecasts; Note: Returns are total returns, except for credit, where we forecast excess returns versus government bonds. Commodity returns are calculated relative to futures to account for carry. Volatility is 10y realized vol. All currency returns are shown as XXXUSD return. *Currency forecasts shown for USDXXX. USDIDR data as of January 15 due to a holiday.

Foreign exchange and commodities reinforce the broader risk-on narrative but with clear sequencing. We expect USD to weaken in 1H26 as US rates converge towards those abroad and global risk appetite improves, before strengthening later in the year as relative US growth reasserts itself. This creates an environment in which carry and risk-sensitive currencies outperform, particularly once the market transitions into a 2H 'carry regime'. In commodities, we maintain a clear preference for metals over oil. Oil markets are characterized by near-term oversupply and rising inventories, with the adjustment occurring largely through curve contango rather than outright price collapse, keeping spot prices supported at lower levels. By contrast, metals benefit from easing monetary policy, demand for real assets and tight inventories, leading Morgan Stanley to hold above-consensus forecasts for most industrial and precious metals. Our commodity strategists are now highlighting the gold bull case for 2H26 (see [Gold: Bull Case in Play](#), January 23, 2026).

Taken together, the cross-asset outlook for 2026 is one of constructive risk-taking, driven less by macro fear and more by growth, earnings and policy normalization.

See [US Economics Weekly: Consumption is overshooting, but we think it can continue](#), January 16, 2026, and [Morgan Stanley Research: Key Forecasts](#), January 20, 2026, for full details.

QIS Performance Outlook – Using a One-Size-Fits-All Framework

The previous two sections set the stage for our outlook for QIS. Effectively, we saw a **positive return from QIS in 2025, particularly since the equity market trough after 'Liberation Day' on April 8, 2025**. At the same time, the **economic environment we expect in 2026 is not too dissimilar to the one in 2H25**: inflation dropping in major economies, rates still on the way lower, an economy that is growing modestly in western countries but remains resilient and expected AI productivity gains and capital deployment being positive for financial markets. In some sense: more of the same. **Thus, at a high level we would expect QIS to continue on its positive trajectory in 2026, but there are important nuances.**

In this section, we use two frameworks which apply a consistent one-size-fits-all evaluation framework across strategies to come up with return forecasts. There is (1) the ATLAS framework, which has a short-to-medium-term forecasting horizon and (2) the economic similarity framework, which has a one-year forecasting horizon. Both are used to arrive at QIS return forecasts and complement each other. For volatility carry, we use our tailored framework introduced in [Timing of Volatility Carry Strategies – Using Signals from the Volatility Surface](#), April 1, 2025, to complement the other two approaches.

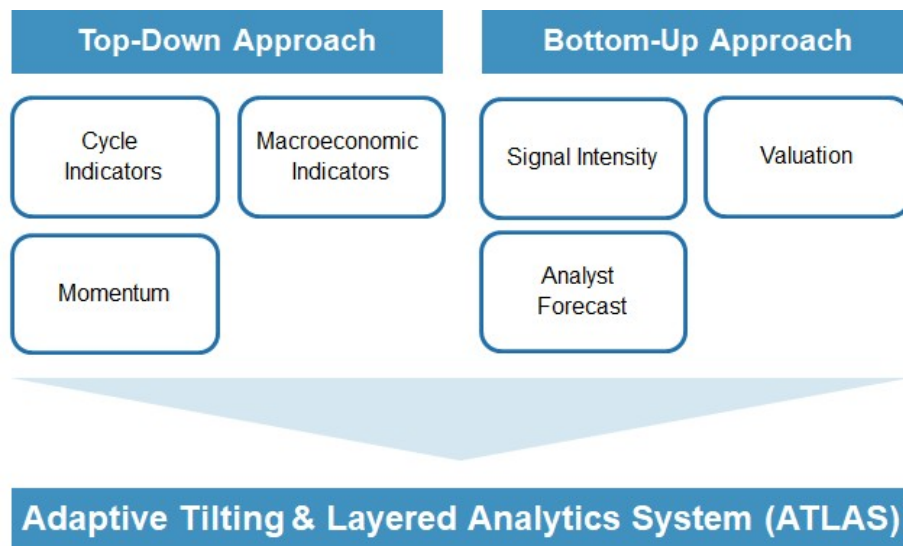
Adaptive Tilting & Layered Analytics System (ATLAS)

Our first port of call for analyzing the QIS return outlook is our multi-dimensional QIS return forecasting framework – **ATLAS (Adaptive Tilting & Layered Analytics System)**. This system combines top-down (i.e., return-based) and bottom-up (i.e., holdings-based) views about the return outlook of a strategy.² ATLAS is a broad, multi-pillar framework to come up with an outlook for QIS returns. The top-down views for each strategy are derived from return outlook-relevant information around cycle, macro and strategy momentum. The bottom-up views for each strategy are informed by signal strength, valuation and analyst expectations. Forecasting returns is done by translating each metric's current reading into a forward return forecast using simple mapping rules (e.g., bucketing into regimes, using percentiles or conditioning on positions), first aggregating forecasts within each category into a single category signal (often with weights tied to confidence or signal strength), and then combining the category signals into an overall score. That final score is used to tilt exposures either through leverage adjustments within a strategy or cross-sectionally across strategies, increasing risk when the combined forecast is favorable and scaling down when it is not.³

Note that we do not recommend to use ATLAS as stand-alone framework but instead in combination with other frameworks to think about QIS expected performance. Put differently, we use ATLAS as input for a quantamental return outlook process.

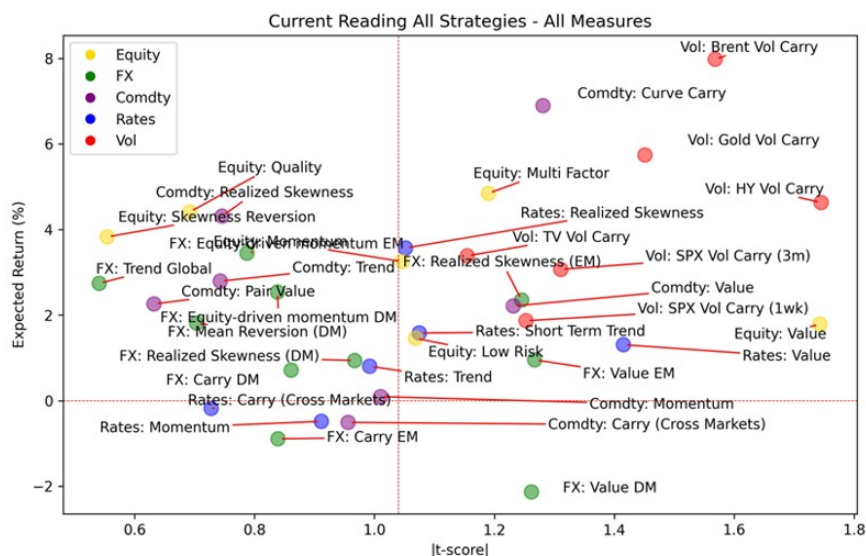
2. ATLAS is a one-month forward (annualized) expected return framework; we interpret the current reading directionality for 2026 and update it through the year. Forecasting power tends to persist for up to three months.

3. If we do not have information from all six metrics (three bottom-up, three top-down), we use whatever information is available from the remaining metrics to come up with a return forecast.

Exhibit 7: The ATLAS framework

Source: QIS Research

We show our return forecasts based on our multi-dimensional QIS ATLAS framework in [Exhibit 8](#). The top-down metrics are available for all strategies in [Exhibit 6](#). The bottom-up metrics are not available for all strategies and we add them to the overall forecast where available. The x-axis gives the conviction of the return expectation and is proxied by the average absolute t-statistic of the underlying indicators. Each indicator's t-stat summarizes the one-month forward strategy return difference versus alternative market environments; we use $|t|$ to weight indicators within each pillar. The y-axis shows expected returns based on the framework with a horizontal line that splits positive from negative return outlooks. Expected returns are one-month forward forecasts (shown annualized) for strategy return series volatility-scaled to a 5% annualized volatility target. It is quite evident that there are not that many strategies with negative expected returns. In fact, we find only for FX value DM a more pronounced negative return expectation. This is quite different, for example, from the situation in the [2025 mid-year outlook](#) where we actually had on average a positive but more muted expectation of QIS returns. Among the strategies that are covered, we can highlight particularly strong return expectations for strategies in the equity and volatility space.

Exhibit 8: ATLAS return readings and conviction by QIS

Source: QIS Research; Note: Data as of January 23, 2026.

We highlight in [Exhibit 9](#) the strategies which are expected to have a particularly high return based on our multi-dimensional forecasting model. The ten strategies which have the highest expected return based on the aggregate are highlighted in red. Not surprisingly, we find that volatility carry strategies appear in this category. Additionally, we find single-name equity strategies – multi factor and quality. On the commodities side it is curve carry that looks rather strong. Finally, our realized skewness strategy appears to show an attractive return outlook, particularly in commodities, equities and rates. This outlook echoes the strategies that did well in 2025 already. Thus, our framework also comes to the conclusion that 2026 is likely to be a continuation of what we saw in 2025.

Exhibit 9: Top ten QIS by outlook dimension

Ranking	Cycle	Momentum	Macro	Overall
1	MSVCM1CO Vol: Brent Vol Carry	MSCFVAE1 FX: Value EM	MSCB2EXB Comdty: Curve Carry	MSVCM1CO Vol: Brent Vol Carry
2	MSCB2EXB Comdty: Curve Carry	MSVCM1CO Vol: Brent Vol Carry	MSVCM1CO Vol: Brent Vol Carry	MSCB2EXB Comdty: Curve Carry
3	MSSKEWB0 Rates: Realized Skewness	MSQEAAGU Equity: Multi Factor	MSVCW1GC Vol: Gold Vol Carry	MSVCW1GC Vol: Gold Vol Carry
4	MSQEAAGU Equity: Momentum	MSQEAAGU Equity: Value	MSCBVMHH Vol: HY Vol Carry	MSQEAAGU Equity: Multi Factor
5	MSVCW1GC Vol: Gold Vol Carry	MSSKEWC0 Comdty: Realized Skewness	MSFVWGTI Vol: TV Vol Carry	MSCBVMHH Vol: HY Vol Carry
6	MSQEAAGU Equity: Quality	MSCBVMHH Vol: HY Vol Carry	MSSKEWE0 Equity: Skewness Reversion	MSQEAAGU Equity: Quality
7	MSSKEWE0 Equity: Skewness Reversion	MSCB2EXB Comdty: Curve Carry	MSQEAAGU Equity: Multi Factor	MSSKEWC0 Comdty: Realized Skewness
8	MSFVWGTI Vol: TV Vol Carry	MSVCW1GC Vol: Gold Vol Carry	MSQEAAGU Equity: Quality	MSSKEWE0 Equity: Skewness Reversion
9	MSSKEWFD FX: Realized Skewness (DM)	MSSKEWFE FX: Realized Skewness (EM)	MSSKEWC0 Comdty: Realized Skewness	MSSKEWB0 Rates: Realized Skewness
10	MSSKEWFE FX: Realized Skewness (EM)	MSQEAAGU Equity: Quality	MSSKEWB0 Rates: Realized Skewness	MSCFME1 FX: Equity-driven momentum EM

Equity
FXComdty
RatesVol
Cross Asset

Source: QIS Research; Note: Data as of January 23, 2026.

Economic similarity

Based on a recent article by Mulliner et al. (2025),⁴ we run an analysis that predicts QIS returns based on identifying periods in the past that are economically similar to the one we expect for 2026. The idea is that the performance of QIS in such similar periods in the past is a good indicator for future performance.

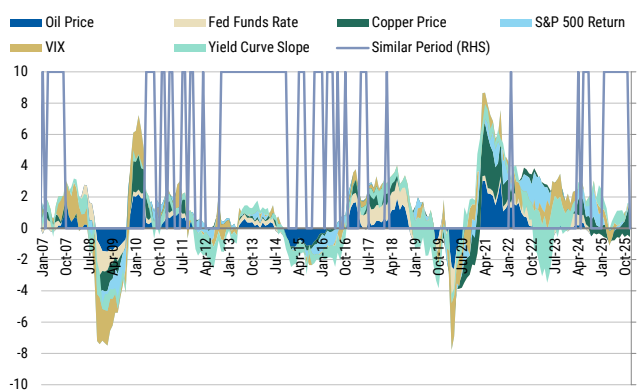
4. Mulliner, Amara, Campbell Harvey, Chao Xia, Ed Fang, and Otto van Hemert (2025): "Regimes." Working Paper, pp. 1-22.

The approach we choose relies on the oil price, the fed funds rate, the copper price, S&P 500 return, VIX and the yield curve slope as indicators to measure similarity. For price-based indicators, we work with annual returns and economic indicators with annual changes. In both cases, we normalize by their 10-year standard deviation. Indicators are winsorized at +/-3 to arrive at the signals s . We then calculate the aggregate signal S as:

$$S = \sqrt{\sum_i (s_i - e_i)^2}$$

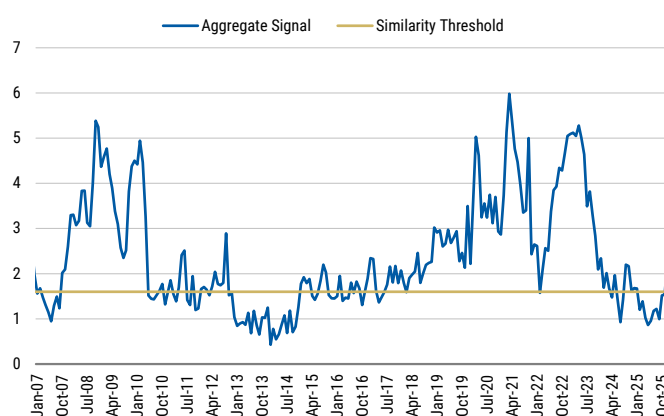
where i stands for the respective indicator and e for the normalized expected economic and market environment based on our fundamental outlook (see [Economic Backdrop](#) for details).⁵ All periods where S is in the lowest quintile are selected as being 'similar' to our expected economic environment. We average the returns of the various representative QIS used in this article during those periods to arrive at a return forecast.

Exhibit 10: Normalized economic indicators (standard deviations)



Source: QIS Research; Note: Data as of December 31, 2025.

Exhibit 11: Aggregate economic similarity indicator (lower number = closer to predicted scenario)



Source: QIS Research; Note: Data as of December 31, 2025.

In [Exhibit 10](#), we see that the 'calmer' market period in 2012, 2013 and 2014 was particularly close to the market dynamics our economists and strategists expect for 2026. The global financial crisis, Covid and 2022 periods are particularly different and the return dynamics of QIS in those periods are less helpful to understand what 2026 may bring. It is also worthwhile to highlight that the indicators have a tendency to be in the same direction in periods of distress, showing that there appears to be a common underlying economic driver. These indicator-based considerations are confirmed when looking at the aggregate signal in [Exhibit 11](#).

The outlook for QIS in 2026 based on their returns in periods with economic similarity further corroborates our narrative of 'more of the same'. In [Exhibit 12](#), we find overall positive returns from commodities, equities and volatility strategies. On the commodity side, curve carry, realized skewness and trend come out as particularly strong. Commodity value is less profitable while we expect pairs value – where we look at economically

5. For VIX we do not have an explicit outlook for 2026. Thus, we assume that it will revert to its long-term average of 18 since 2010.

integrated pairs – to do well. In single equities, a multi-strategy approach as well as momentum and quality have a positive outlook while returns for low risk and value are expected to be up but less strongly. We also expect investing based on realized skewness of equity index returns to be a profitable strategy. Volatility carry is overall strong although the signals are a bit mixed for S&P 500 short volatility.

Exhibit 12: Performance forecast for selected QIS

Asset Class	Strategy	Expected Performance	Asset Class	Strategy	Expected Performance
Comdty	Carry (Cross Markets)	-0.9%	FX	Carry DM	-1.7%
	Curve Carry	9.5%		Carry EM	1.6%
	Momentum	0.2%		DM Reversion	4.7%
	Pair Value	6.3%		Equity-driven momentum DM	8.1%
	Realized Skewness	5.6%		Equity-driven momentum EM	4.7%
	Trend	2.9%		Realized Skewness (DM)	-1.2%
	Value	-1.1%		Realized Skewness (EM)	-2.4%
Equity	Low Risk	4.6%		Trend Global	-1.0%
	Momentum	7.9%		Value DM	4.1%
	Multi Factor	11.6%		Value EM	3.7%
	Quality	7.2%	Rates	Carry (Cross Markets)	-1.1%
	Realized Skewness	10.4%		Momentum	-3.6%
	Value	2.3%		Realized Skewness	7.6%
Vol	Brent	6.1%		Short Rates Trend	-0.2%
	Credit HY	7.4%		Trend	0.0%
	SPX Vol Carry (1wk)	0.1%		Value	-2.2%
	SPX Vol Carry (3m)	-1.8%	Cross	Skewness	9.9%
	SPX Vol Carry (Upside Vol)	7.0%		Asset Trend	6.5%
	TV Vol Carry	10.3%	Mean		3.8%
	VIX	11.9%	Median		4.4%

Source: QIS Research; Note: Data as of December 31, 2025.

In the FX space, mean reversion and equity-driven momentum stand out as strategies that we expect to perform well. We also expect FX value to deliver positive returns in emerging and developed markets. The remaining set of FX strategies is flat to down. On the rates side this analysis shows that styles might be flat to down in 2026 with the exception of realized skewness.

In the cross-asset space we find positive returns for a cross-asset realized skewness strategy. Trend following is positive. As mentioned already, this trend outlook will be complemented by a deeper analysis of market dynamics, which are supportive for trend-following returns in 2026.

When we take the average across the return expectations from all strategies, we arrive at a positive average expectation of 3.8%. The median expected return is also positive at 4.4%, highlighting that the return outlook is reasonably balanced and not dominated by a few outsized predictions. Thus, deriving QIS return expectations from periods with economic similarity supports our generally favorable outlook for QIS. That said, geopolitical uncertainty is elevated, which the economic similarity-based approach does not capture well.

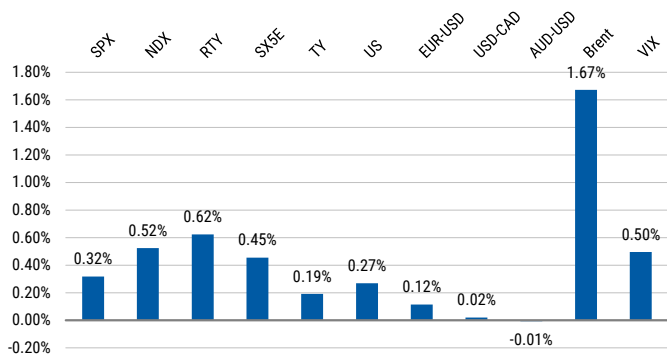
Volatility-focused framework

Next, we use our volatility-focused framework from [Timing of Volatility Carry Strategies](#), April 1, 2025, to derive an outlook for volatility carry strategies.

The framework provides us with some interesting results: we expect pretty much all strategies to perform in line with long-term returns ([Exhibit 13](#) and [Exhibit 14](#)). Thus, we should see a rather attractive performance coming out of the volatility sector this year. Based on these readings, we would not recommend a large deviation from long-term strategic allocations among volatility carry strategies. Particularly, this means a **potential underweighting of FX volatility carry** compared to other asset classes due to a generally lower return expectation.

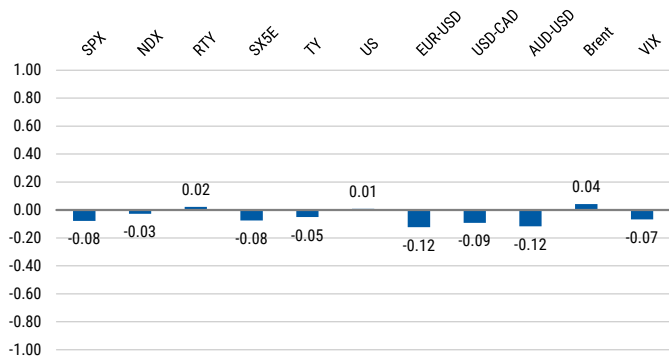
We will contrast these readings later in [QIS Performance Outlook – Fundamental Analysis](#) with a more fundamental approach looking at links between our strategy returns, volatility term structure and changes thereof over time.

Exhibit 13: Volatility-specific monthly return forecast



Source: QIS Research; Note: Data as of January 21, 2026.

Exhibit 14: Volatility-specific Z-scores



Source: QIS Research; Note: Data as of January 21, 2026.

QIS Performance Outlook – Fundamental Analysis

This section is mainly concerned with fundamental analyses complementing the historical top-down analyses we have done in previous sections. We focus in our analysis on providing additional considerations around the trend, equity and volatility QIS outlook. We establish if returns of the respective QIS are likely to follow historical patterns or if the adage 'this time it is different' holds.

Trend

We like trend following as a strategy for 2026 based on its inherent convexity as well as strong return expectations. When we look at the performance in the long term it was the bond and rates markets that drove the performance ([Exhibit 15](#)). However, since the high watermark of our trend-following strategy, rates and bond markets were actually detractors from performance, driven partially by speculation around monetary policy.

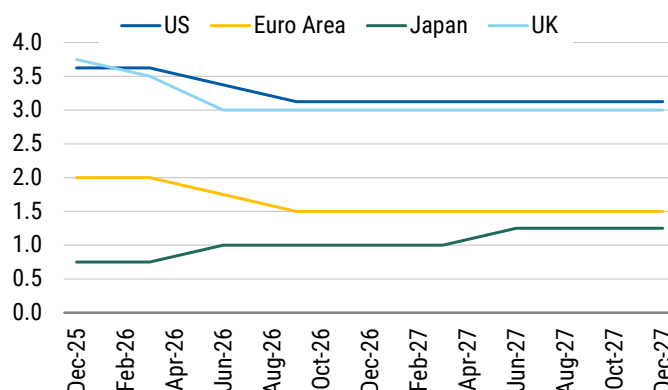
In 2026, we think that the fortunes of rates trend will change for two reasons. For one, we still see monetary policy rates at the front end coming down, leading to possible rates trends that can be captured ([Exhibit 16](#)). Second, the lower rates at the front end mean that the yield curve is expected to steepen in many regions (e.g., 93bp in the US between fed funds and 10-year govies). This steepening would provide potentially additional income in the rates space as trend may lean in the medium term into the carry earned from government futures. With these dynamics, headwinds to trend-following strategies should abate and turn into positive tailwinds. However, in equities, we expect more of the same by equity markets continuing to go up (see, for example, [Exhibit 6](#)), driving positive trend-following returns. This overall positive picture is also supported by the 10.6% the strategy earned between June 20, 2025 and January 9, 2026, illustrating strong positive momentum for enhanced trend following in the current market environment.

Exhibit 15: Performance decomposition by asset class
(enhanced trend)

Asset Class	Long term	Since High Water Mark	Since April 30 2025
Bonds	230.7%	-7.2%	-1.3%
Short Term Rates	201.2%	-6.1%	-1.7%
Credit	95.4%	6.9%	3.3%
Equity	95.3%	3.6%	6.0%
Commodities	92.6%	6.5%	-0.8%
FX	34.8%	-4.0%	1.1%

Source: QIS Research; Note: Data as of January 9, 2026.

Exhibit 16: Rates forecasts for quarter ends



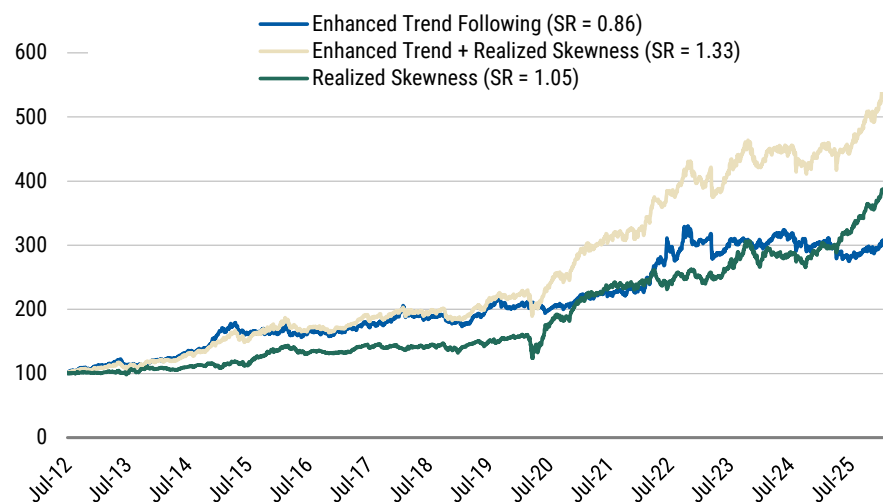
Source: Morgan Stanley Research forecasts; Note: Data as of January 23, 2026.

We recently introduced our new realized skewness strategy in [Quantifying Asymmetry – Realized Skewness as a Core Portfolio Strategy](#), December 3, 2025. We note that this realized skewness strategy combines nicely with trend following as it has a tendency to pick up mean reversion after market drawdowns, which is the market scenario when trend

following tends to give up returns. In [Exhibit 17](#), we show that the Sharpe ratio increases from 0.86 in enhanced trend following alone to 1.33 when combining it with realized skewness. With the latter already having had a great return in 2025 and a strong outlook in its own right, we think that a **combination of enhanced trend following with realized skewness should be a core holding in a QIS portfolio**. The addition of intraday trend-following strategies can provide additional diversification by being able to pick up strong returns in periods of adverse market shocks.

These considerations together with the strong return expectations expressed in our ATLAS and economic similarity frameworks prompt us to be constructive on enhanced trend following as well as a combination thereof with realized skewness.

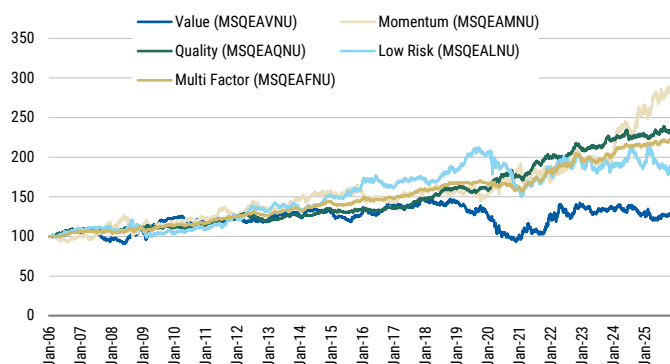
Exhibit 17: Enhanced trend combines well with cross-asset skewness



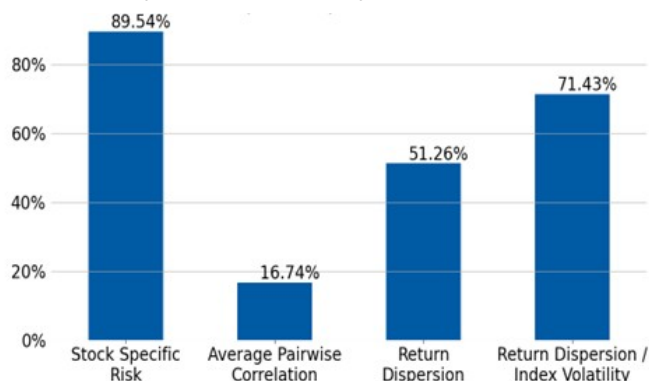
Source: QIS Research; Note: Data as of January 23, 2026.

Equity

In previous sections we showed a generally supportive outlook for single equity strategies. When we dive a bit deeper, this observation continues to hold. First, we show returns for equity systematic strategies in the US in [Exhibit 18](#). Momentum is particularly strong over the past few years, followed by quality. Less strong are returns in value and low risk, which results overall in strong multi-factor returns. Given that the market environment in 2026 is likely to be a continuation of some of the characteristics we saw in 2025, this supports a positive systematic equity outlook.

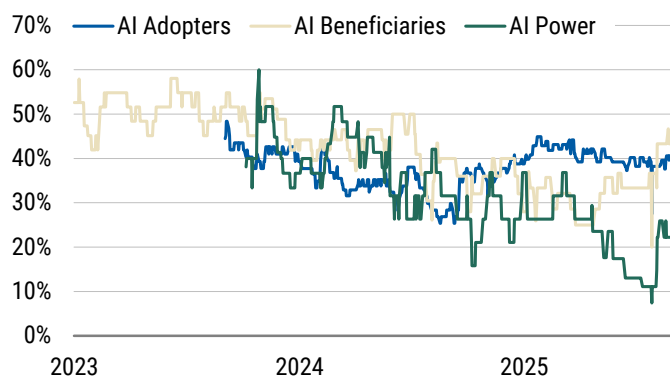
Exhibit 18: Equity QIS US performance

Source: QIS Research; Note: Data as of January 9, 2026.

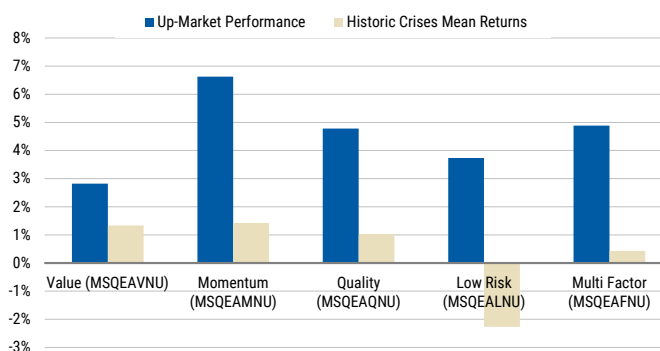
Exhibit 19: Key indicators for equity QIS

Source: QIS Research; Note: Data as of December 31, 2025.

Indicators which we find helpful to characterize the opportunity set of systematic equity strategies focus on stock-specific risk, pairwise correlations among equities and return dispersion. We highlight the respective metrics for US equities in [Exhibit 19](#). Stock-specific risk is at the 90th percentile and average pairwise correlation at the 17th. Both indicators point towards an equity market that provides lots of alpha opportunities rather than being driven only by macro risk. Return dispersion is the only metric that is around its median, providing no particular support for equity systematic strategy outperformance. That said, when normalizing this metric by index volatility we get an attractive score in the 71st percentile, which is also supportive for alpha generation. In short, equity markets in the US show a lot of idiosyncrasy, which tends to be positive for systematic equity strategies.

Exhibit 20: Overlap of visionary quality with AI baskets

Source: QIS Research; Note: Data as of December 31, 2025.

Exhibit 21: Performance of equity strategies by market phase

Source: QIS Research; Note: Data as of December 31, 2025.

It is worthwhile to outline how our definition of quality combines traditional, more defensive aspects of quality with a more growth-oriented angle. Essentially, we leverage our work on R&D signals and the identification of top innovators (see [here](#) and [here](#)) to extend traditional quality by a growth-oriented dimension. Overall, this provides a performance boost to quality but it also leads to a pick-up of some AI-related tilts in the allocation. As can be seen in [Exhibit 20](#), we find that currently 20-50% of the names in selected AI-related baskets also show up as long holdings of our visionary quality strategy

(see [Thematic Tides: Quantifying the Next Big Waves](#), November 10, 2025, for an overview of our thematic strategies and portfolio construction). Given that Morgan Stanley expects the overall AI theme to continue to drive investment and market returns, this pick-up should provide an interesting additional performance boost to our visionary quality strategy.

The final argument prompting us to be positive on systematic equity strategies is captured when studying the conditional returns of the strategies in [Exhibit 21](#). In up markets all styles tend to produce positive average returns. Given that we expect the S&P 500 to reach 7,800 in our base case, this should spell out positive systematic equity returns. However, should the bear case materialize where we would see an end-2026 value of the S&P 500 at 5,600, we would see flatter but still positive returns in the multi-factor strategy. Thus, equity factors are well suited to be part of a portfolio that is geared towards doing well in our base case scenario but will remain resilient even in the adverse bear case.

Thus, based on past performance, alpha opportunities in individual equities, a positive but limited exposure to the AI theme and a strong scenario-oriented return profile, systematic equity strategies should do well overall. For investors interested in exposure to the AI theme, our visionary quality factor provides an interesting alternative.

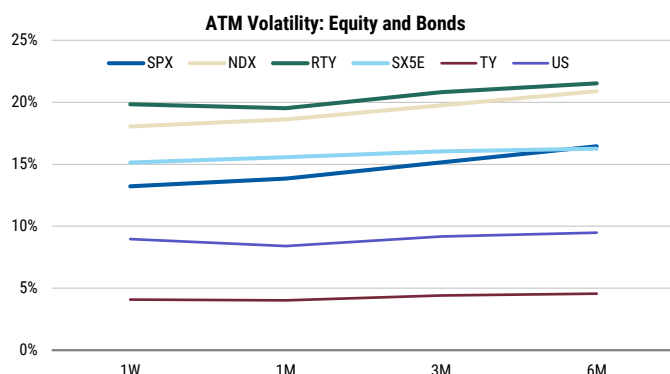
Volatility

Fundamental considerations

Similar to the case of equity strategies, we find that our ATLAS framework and the comparison to similar economic periods confirm an overall positive outlook for volatility carry strategies. Given that these frameworks have a medium-term outlook over 2026 based mainly on economic dynamics, we now add a more fundamental, bottom-up angle focused on linking volatility market fundamentals with an outlook for volatility carry.

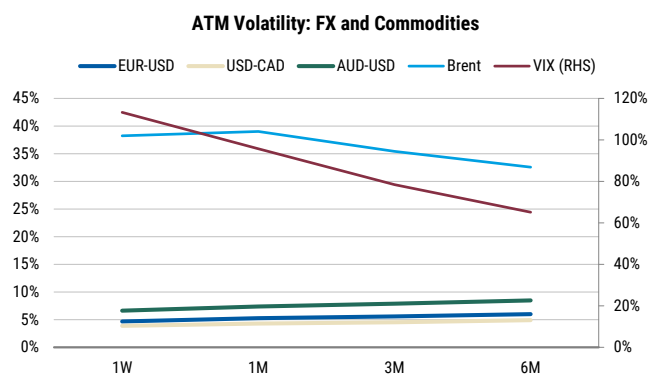
Our approach is focused on linking volatility market and strategy return over the past year to infer what we may see in terms of 2026 performance as we expect somewhat similar market dynamics. We also analyze the current market dynamics to establish if we currently see a good entry point in volatility carry strategies.

Exhibit 22: ATM volatility: Equity and bonds



Source: QIS Research; Note: Data as of January 21, 2026.

Exhibit 23: ATM volatility: FX and commodities



Source: QIS Research; Note: Data as of January 21, 2026.

When we approach the outlook of volatility carry more fundamentally, we are interested in the level of the volatility risk premium, as this is a relevant forecaster for the profitability of volatility carry strategies. We calculate the volatility risk premium as the implied versus realized volatility spread (IV-RV).⁶ We are interested in this spread over the past year as it provides a good indication of what we can expect in the coming year, given our hypothesis of broadly comparable market dynamics. Looking at the median IV-RV metric over the past six and 12 months, we find consistent results: the volatility risk premium is in most markets above the long-term median ([Exhibit 24](#)). This provides a supportive view, indicating that volatility carry should deliver at least a performance in line with historical returns.

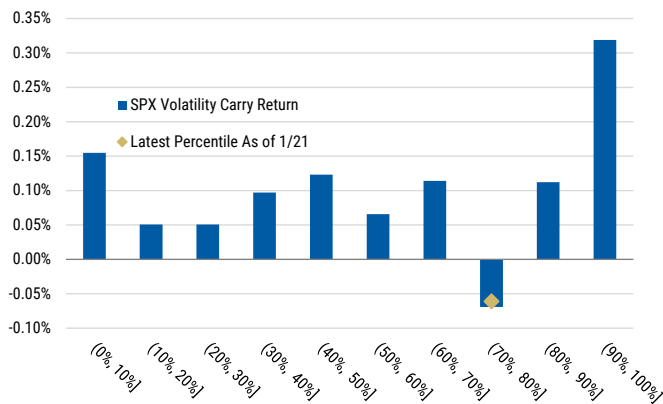
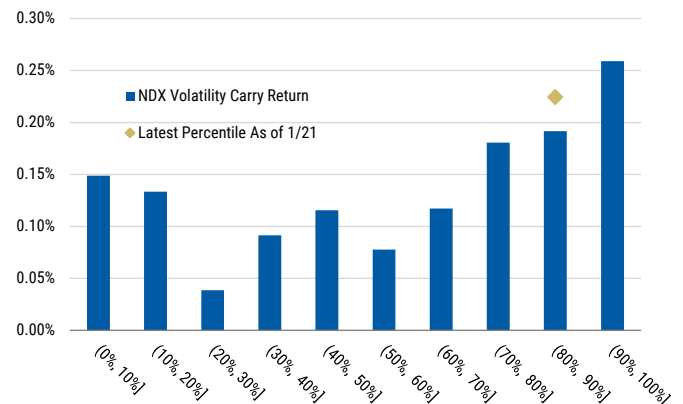
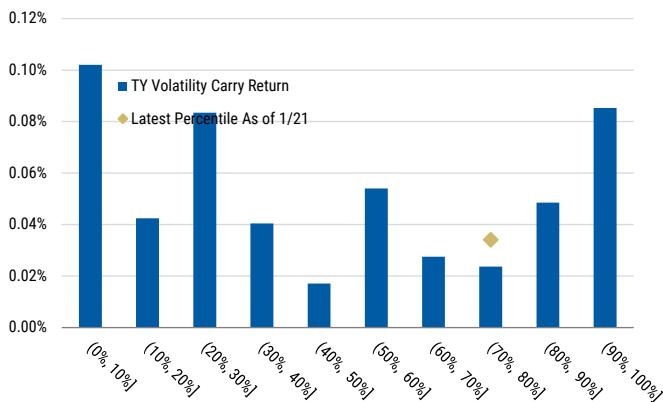
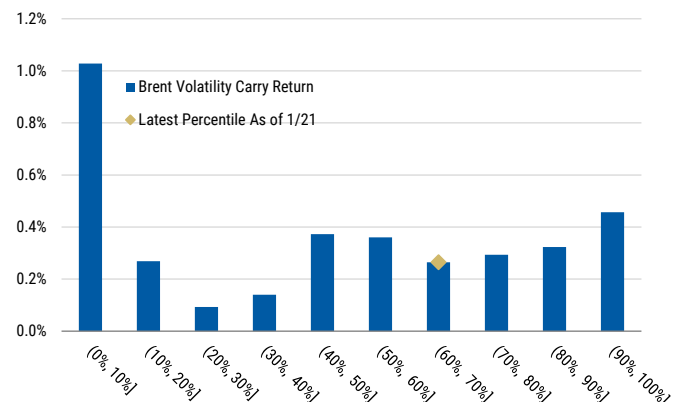
Exhibit 24: Implied versus realized volatilities over various time horizons

		SPX	NDX	RTY	SX5E	TY	US	EUR-USD	USD-CAD	AUD-USD	Brent
		1W	1W	1W	1W	1W	1W	1W	1W	3M	1M
Since 2012		0.59%	0.46%	1.43%	1.11%	0.36%	0.49%	0.37%	0.33%	0.34%	2.88%
12 Month	Median	0.72%	0.71%	1.83%	1.23%	0.70%	1.42%	0.40%	0.18%	0.78%	4.10%
	Percentile	53.99%	53.70%	56.02%	51.97%	67.37%	74.33%	52.35%	45.28%	61.85%	61.59%
6 Month	Median	0.85%	1.10%	0.92%	1.16%	0.95%	1.82%	0.52%	0.16%	0.89%	4.63%
	Percentile	55.73%	56.99%	46.54%	51.00%	78.01%	81.74%	56.25%	44.22%	64.30%	64.06%
Current (21-Jan-2026)	Level	2.66%	5.71%	4.85%	3.94%	1.02%	2.14%	0.75%	0.79%	1.97%	6.20%
	Percentile	74.91%	89.78%	82.31%	77.91%	80.14%	86.22%	62.90%	66.36%	83.93%	71.92%

Source: QIS Research; Note: Data as of January 21, 2026.

It is important to outline that the forecasting power of IV-RV is not linear. In fact, this metric has forecasting power for volatility carry strategies predominantly in the extremes. In equities it is mainly the top decile of IV-RV where we see strong forward returns. We show selected markets in [Exhibit 25](#) to [Exhibit 28](#). The diamond shape in those graphs illustrates where IV-RV is currently. For the Nasdaq current IV-RV levels indeed suggest an elevated return expectation. However, we would expect levels over 2026 to be more in line with averages over the past six and 12 months, which would point towards a Nasdaq volatility carry performance in line with historical values. We find similar results for Treasury and oil markets ([Exhibit 27](#) and [Exhibit 28](#)) – it's at the extremes of IV-RV percentiles where we see larger effects on expected performance and neither six (or 12) month nor current levels are there. Thus, overall we expect volatility carry returns in line with historical averages

6. Realized volatility is calculated over a rolling 20-day window. Implied volatility is at the money unless otherwise noted.

Exhibit 25: 1W forward return by IV-RV percentile: SPX**Exhibit 26:** 1W forward return by IV-RV percentile: NDX**Exhibit 27:** 1W forward return by IV-RV percentile: TY**Exhibit 28:** 1W forward return by IV-RV percentile: Brent

When we look at current levels for implied volatility, term structure and IV-RV, we find generally supportive levels ([Exhibit 29](#)). IV-RV are above the median across markets. Implied volatilities are above the median in US equities and Brent but quite low in FX volatility markets. While these numbers fluctuate a lot, the readings are an indication that we currently have a reasonably attractive entry point for volatility carry – be that for shorter-term trades or longer-term QIS allocations.

Exhibit 29: Current implied volatility slope, levels and implied versus realized across markets

	SPX	NDX	RTY	SX5E	TY	US	EUR-USD	USD-CAD	AUD-USD	Brent
ATM Implied Volatility										
Tenor	1W	1W	1W	1W	1W	1W	1W	1W	1W	3M
Level	13.22%	18.04%	19.83%	15.14%	4.08%	8.97%	4.67%	3.88%	6.63%	35.44%
Percentile	56.93%	55.29%	47.42%	34.26%	24.64%	39.11%	6.08%	2.86%	7.89%	64.07%
Implied Volatility Slope										
Tenor	1M-1W	1M-1W	1M-1W	1M-1W	1M-1W	1M-1W	1M-1W	1M-1W	1M-1W	6M-3M
Level	0.62%	0.57%	-0.31%	0.44%	-0.06%	-0.57%	0.61%	0.41%	0.75%	-2.87%
Percentile	48.86%	45.86%	32.90%	67.43%	30.09%	11.66%	76.78%	69.52%	80.26%	13.10%
Implied-Realized Volatility										
Tenor	1W	1W	1W	1W	1W	1W	1W	1W	1W	3M
Level	2.66%	5.71%	4.85%	3.94%	0.99%	2.07%	0.79%	0.81%	1.95%	6.20%
Percentile	74.56%	88.66%	80.44%	73.90%	79.21%	84.51%	62.31%	70.72%	85.54%	65.07%

Source: QIS Research; Note: Data as of January 21, 2026.

We also turn to our analysis of economically similar periods from [QIS Performance Outlook – Using a One-Size-Fits-All Framework](#) to assess if IV-RV is in line with the levels we saw in the past. Overall, we find similar levels in equities IV-RV today when compared to similar periods in the past. In rates and Brent IV-RV appears currently actually higher compared to similar periods in the past ([Exhibit 30](#)). Overall, though, this analysis also point towards robust volatility carry strategy performance.

Exhibit 30: Expectations for IV-RV given the market environment

Period	Metric	SPX	NDX	RTY	SX5E	TY	US	EUR-USD	USD-CAD	AUD-USD	Brent
Similar Periods	IV - RV	0.53%	0.08%	0.55%	1.30%	0.41%	0.48%	0.62%	0.42%	0.04%	2.97%
	Realized Returns (@6% Volatility)	6.4%	4.6%	5.9%	7.8%	9.2%	6.6%	7.6%	5.6%	-1.0%	6.6%
Less Similar	IV - RV	-0.02%	-0.15%	1.04%	1.16%	0.26%	0.41%	0.51%	0.39%	0.33%	1.11%
	Realized Returns (@6% Volatility)	4.9%	7.3%	6.3%	8.9%	6.2%	6.5%	6.6%	2.8%	3.1%	14.7%
Current		2.66%	5.71%	4.85%	3.94%	0.99%	2.07%	0.79%	0.81%	1.95%	6.20%
6 Month Median	IV - RV	0.85%	1.10%	0.92%	1.16%	0.95%	1.82%	0.52%	0.16%	0.89%	4.63%

Source: Morgan Stanley Research; Note: Data as of January 21, 2026.

The one thing that could speak against this analysis is the possibility for more or larger jumps in financial market prices as a consequence of (geo-) political developments. This may impact the returns to volatility carry negatively. However, when we estimate jump intensities through a jump-diffusion process we do not find a substantially and persistently elevated tendency for price jumps in equity and other markets. Another indicator, volatility of realized volatility, is around long-term averages in most markets. Only in equities is this indicator modestly elevated, reaching values seen in 2012 or 2019, well below 2020 levels. Thus, our conclusion is that there is little reliable evidence for persistently increased price jump risks in financial markets that would adversely impact our conclusions so far.

In summary, the volatility risk premium is currently above the median in most markets, indicating a good entry point into volatility carry. Overall, we saw the volatility risk premium at attractive levels across markets over the past year, which should indicate similar levels for the coming year and as a result attractive returns for volatility carry. The volatility risk premium does not reach extreme levels that would result in significantly elevated volatility carry returns. **On the back of our analyses, we think that volatility carry will show a performance roughly in line with long-term averages.** See [Exhibit 36](#) to [Exhibit 46](#) in the [Appendix](#) for more details.

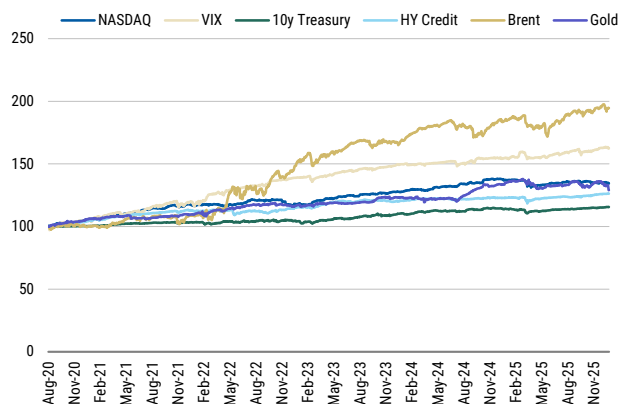
However, we acknowledge that implied volatility and the volatility risk premium move fast and the conclusions of the analyses in this section may shift swiftly.

Portfolio considerations

Having found a generally supportive environment for volatility carry, we focus on the construction of a multi-strategy volatility carry portfolio. Equity volatility carry is a building block that belongs into such a portfolio due to long-term strong returns and a market environment that suggests expected returns in line with historical performance. A positive beta loading of equity volatility carry on underlying US equity market returns provides further support given that our strategists expect a strong US equity market return. Among the covered equity markets, Nasdaq volatility carry appears the most attractive given a comparably high volatility risk premium and strong historical returns.

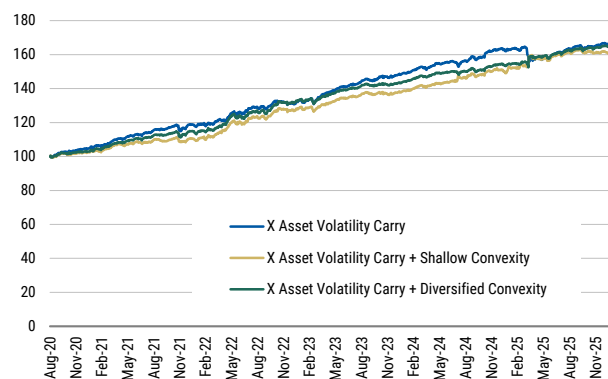
Political uncertainty and downside risks to our equity market forecasts prompts us to **favor out-of-the-money call selling**, which has less exposure to equity market gap down moves and tends to deliver stronger risk-adjusted returns. Demand for upside calls from investors provides further support for this rationale.

Exhibit 31: Selected volatility carry QIS



Source: QIS Research; Note: Data as of January 21, 2026.

Exhibit 32: Broad portfolio of volatility carry QIS



Source: QIS Research; Note: Data as of January 21, 2026.

Given the current economic and political uncertainty, some investors may want to combine Nasdaq upside volatility carry with holding some outright long volatility/protection for further diversification. Given our outlook, the key risk here is smaller gaps down in a generally upwards drifting market. This possibility of shorter, abrupt dips on the back of geopolitical or economic events makes us **favor reactive long gamma strategies as one way to diversify against fast but limited down moves**. Close-to-the-money puts and calls at the front end of the curve are attractively priced for long gamma implementations, delivering strong returns in down gaps. This selection of long options in a gamma strategy is quite different to the shorter-term out-of-the-money call selling we prefer for volatility carry generation in Nasdaq. Thus, preferred volatility carry and long gamma strategy implementations are natural complements. Outright long puts are possibly less attractive, in our view, given pricing and generally positive expectations about equity market returns. Other interesting approaches are long VIX call spread replication as well as dispersion.

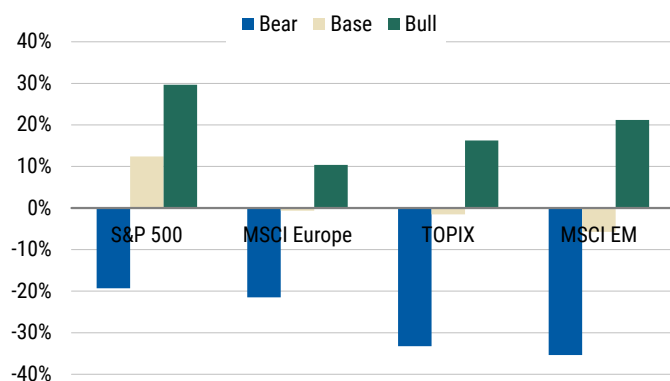
We find similar parameters for rates, credit and commodities as well: over the past year volatility levels and risk premia were not extreme and rather close to historical averages.

Strategies delivered attractive returns at these levels and we would expect this movement to continue. Current levels of volatility risk premia are also around long-term averages, providing a good entry point. Our conviction in volatility carry in FX is somewhat lower, essentially prompting us to favor a broadly diversified portfolio of baseline volatility carry strategies in equities, VIX, credit, commodities and rates (see [Exhibit 31](#) for individual strategy performance and [Exhibit 32](#) for the aggregate).

X Asset Volatility Carry + Shallow Convexity in [Exhibit 32](#) shows the impact of combining a diversified portfolio of volatility carry strategies with shallow gap down protection, which would have helped to buffer volatility carry losses in April 2025.

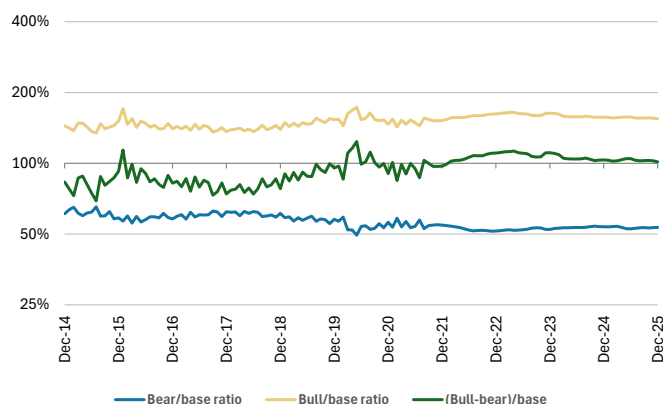
While not part of our base case, we need to acknowledge that there are substantial risks to the economic outlook (our S&P 500 bear case sees the index at 5,600; [Exhibit 6](#)). We illustrate 2026 bull, bear and base case for selected markets in [Exhibit 33](#) to show the degree of uncertainty our strategists build into their forecasts. In [Exhibit 34](#) we show historical bottom-up bull and bear case spreads, which highlights that uncertainty is elevated today compared to historical levels (US equity downside skewness bear versus base at 53% today versus a maximum of 65% in 2015). However, there are regional differences, where we find this downside skewness more pronounced in the US than in European markets. Investors in volatility carry may want to take these elevated downside risks into account when building portfolios.

Exhibit 33: Morgan Stanley bull, bear and base case for equity markets



Source: Morgan Stanley Research forecasts; Note: Data as of January 23, 2026.

Exhibit 34: Morgan Stanley bottom-up bull, bear and base case history for the US



Source: Morgan Stanley Research forecasts; Note: Data as of January 21, 2026.

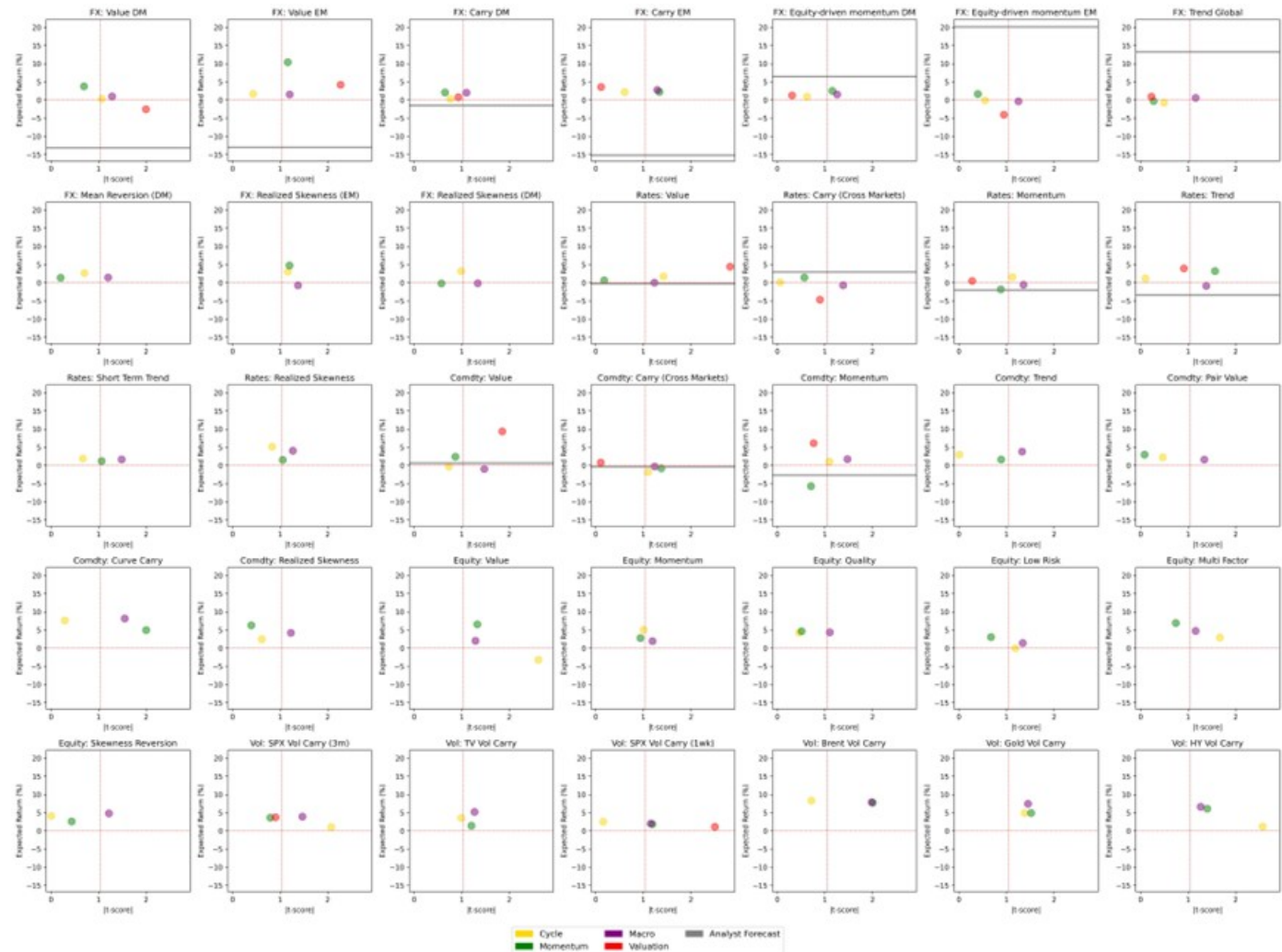
Investors who want to get exposure to volatility carry while protecting against shorter dips as well as the deeper corrections embedded in our bear cases could combine the diversified volatility carry portfolio with long gamma, dispersion and a conditional down variance strategies against deeper corrections (see X Asset Volatility Carry + Diversified Convexity in [Exhibit 32](#)).

Our final thoughts on strategies in the options space evolve around gold. Fundamentally we see gold at 4,800 in our base case, which is around current levels, with our commodity strategists now highlighting the bull case. Gold may provide further upside in case of market distress. The gold option market provides an opportunity given this backdrop. Combining short strangles on gold with a long call spread can provide enhanced gold beta that should fare well in limited upside situations.

Appendix

Overview of ATLAS forecasts by subsignal

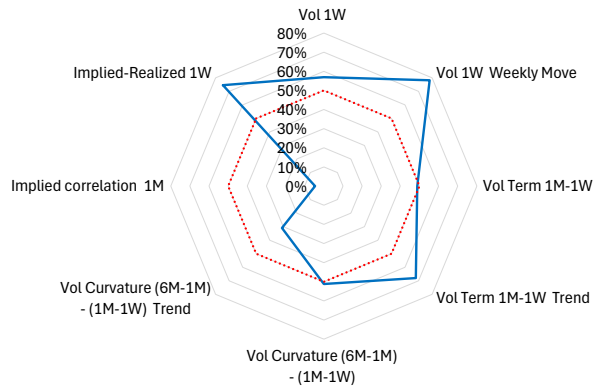
Exhibit 35: ATLAS components by strategy



Source: QIS Research; Note: Data as of December 31, 2025.

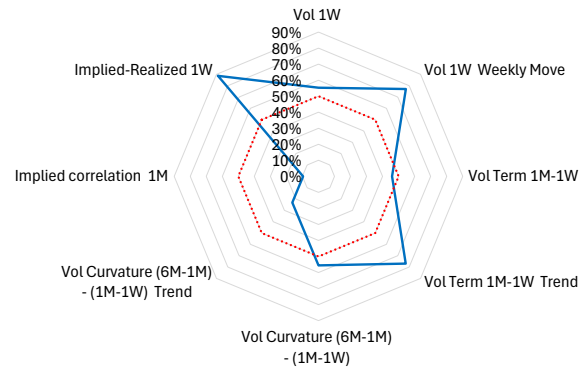
Volatility carry forecast percentiles by signal

Exhibit 36: SPX signal percentiles



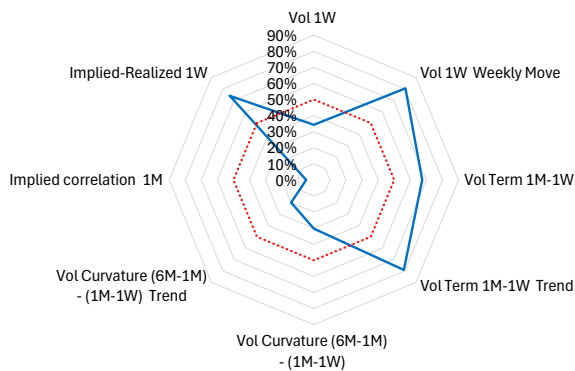
Source: QIS Research; Note: Data as of January 21, 2026.

Exhibit 37: NDX signal percentiles



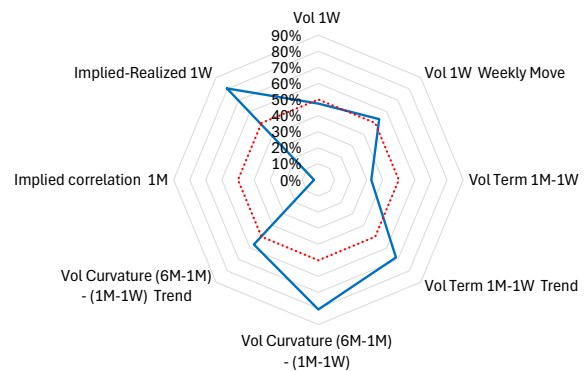
Source: QIS Research; Note: Data as of January 21, 2026.

Exhibit 38: SX5E signal percentiles



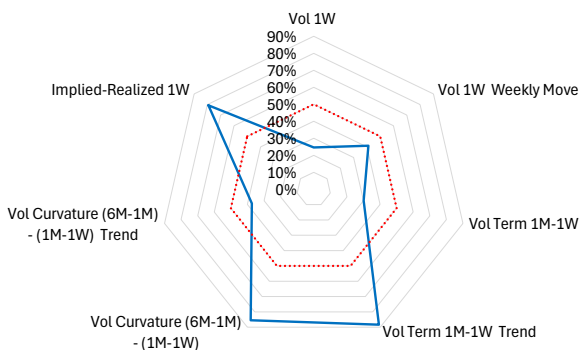
Source: QIS Research; Note: Data as of January 21, 2026.

Exhibit 39: RTY signal percentiles



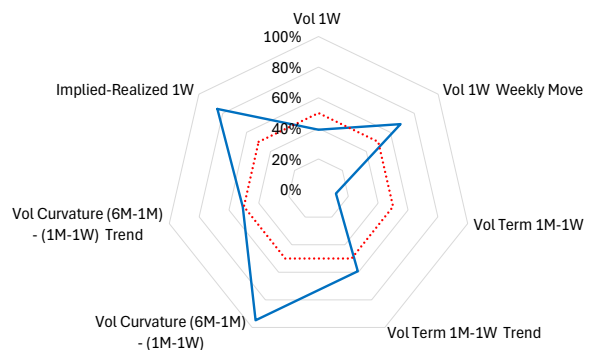
Source: QIS Research; Note: Data as of January 21, 2026.

Exhibit 40: TY signal percentiles

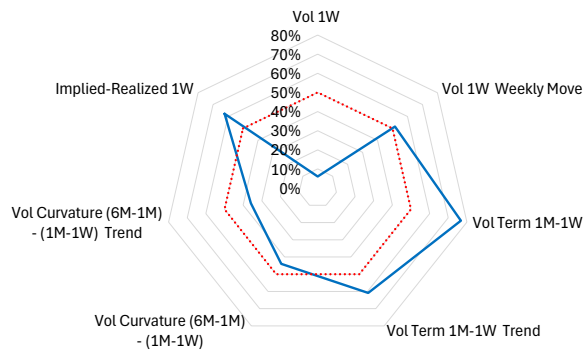


Source: QIS Research; Note: Data as of January 21, 2026.

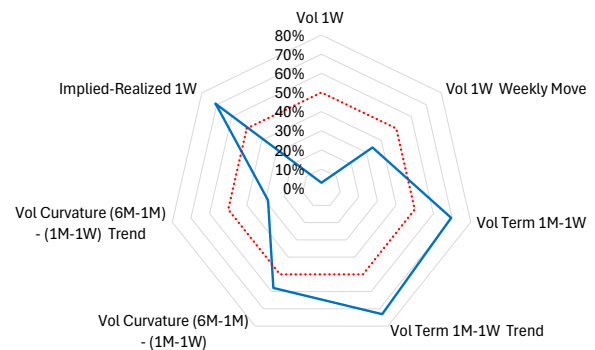
Exhibit 41: US signal percentiles



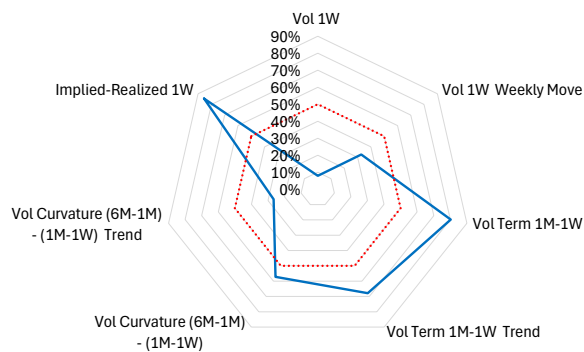
Source: QIS Research; Note: Data as of January 21, 2026..

Exhibit 42: EUR-USD signal percentiles

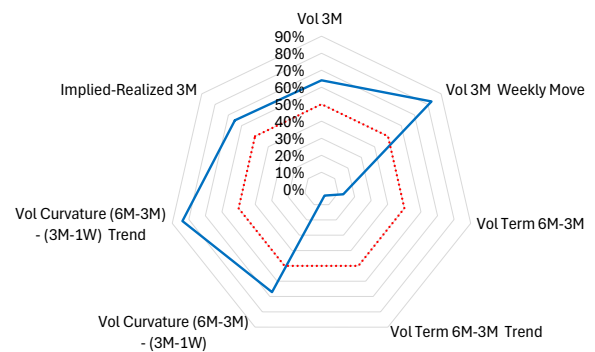
Source: QIS Research; Note: Data as of January 21, 2026.

Exhibit 43: USD-CAD signal percentiles

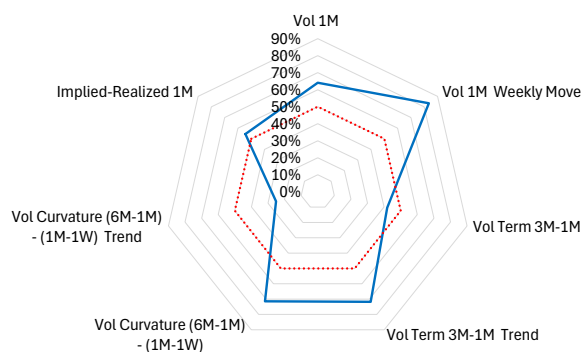
Source: QIS Research; Note: Data as of January 21, 2026.

Exhibit 44: AUD-USD signal percentiles

Source: QIS Research; Note: Data as of January 21, 2026.

Exhibit 45: Brent signal percentiles

Source: QIS Research; Note: Data as of January 21, 2026.

Exhibit 46: VIX signal percentiles

Source: QIS Research; Note: Data as of January 21, 2026.

Appendix: Strategy Risk Factors

Buying calls or call spreads: Investors who buy call options risk loss of the entire premium paid if the underlying security finishes below the strike price at expiration. Investors who buy call spreads (buy a call and sell a further OTM call) also have a maximum loss of the entire up-front premium paid. The maximum gain from buying call spreads is the difference between the strike prices, less the upfront premium paid.

Buying puts or put spreads: Investors who buy put options risk loss of the entire premium paid if the underlying security finishes above the strike price at expiration. Investors who buy put spreads (buy a put and sell a further OTM put) also have a maximum loss of the upfront premium paid. The maximum gain from buying put spreads is the difference between the strike prices, less the upfront premium paid.

Selling calls: Investors who sell covered calls (own the underlying security and sell a call) risk limiting their upside to the strike price plus the upfront premium received and may have their security called away if the security price exceeds the strike price of the short call. Additionally, the investor has full downside exposure that is only partially offset by the upfront premium taken in. Investors short naked calls (i.e. sold calls but don't hold underlying security) risk unlimited losses of security price less strike price. Investors who sell naked call spreads (i.e. sell a call and buy a farther out-of-the-money call with no underlying security position) have a maximum loss of the difference between the long call strike and the short call strike, less the upfront premium taken in, if the underlying security finishes above the long call strike at expiration. The maximum gain is the upfront premium taken in, if the security finishes below the short call strike at expiration.

Selling puts: Put sellers commit to buying the underlying security at the strike price in the event the security falls below the strike price. The maximum loss is the full strike price less the premium received for selling the put. Put sellers who are also long a lower dollar-strike put face a maximum loss of the difference between the long and short put strikes less the options premium received.

Buying strangles: The maximum loss is the entire premium paid (put + call), if the security finishes between the put strike and the call strike at expiration.

Selling strangles or straddles: Investors who are long a security and short a strangle or straddle risk capping their upside in the security to the strike price of the call that is sold plus the upfront premium received. Additionally, if the security trades below the strike price of the short put, the investor risks losing the difference between the strike price and the security price (less the value of the premium received) on the short put and will also experience losses in the security position if he owns shares. The maximum potential loss is the full value of the strike price (less the value of the premium received) plus losses on the long security position. Investors who are short naked strangles or straddles have unlimited potential loss since if the security trades above the call strike price, the investor risks losing the difference between the strike price and the security price (less the value of the premium received) on the short call. Additionally, they are obligated to buy the security at the put strike price (less upfront premium received) if the security finishes below the put strike price at expiration. Strangle/straddle sellers risk assignment on short put positions that become in the money. Additionally, they risk having stock called away from short call positions that become in the money.

Options Disclaimer

Options are not for everyone. Before engaging in the purchasing or writing of options, investors should understand the nature and extent of their rights and obligations and be aware of the risks involved, including the risks pertaining to the business and financial condition of the issuer and the underlying stock. A secondary market may not exist for these securities. For customers of Morgan Stanley & Co. Incorporated who are purchasing or writing exchange-traded options, your attention is called to the publication "Characteristics and Risks of Standardized Options;" in particular, the statement entitled "Risks of Option Writers." That publication, which you should have read and understood prior to investing in options, can be viewed on the Web at the following address: <http://www.optionsclearing.com/about/publications/character-risks.jsp>. Spreading may also entail substantial commissions, because it involves at least twice the number of contracts as a long or short position and because spreads are almost invariably closed out prior to expiration. Potential investors should be advised that the tax treatment applicable to spread transactions should be carefully reviewed prior to entering into any transaction. Also, it should be pointed out that while the investor who engages in spread transactions may be reducing risk, he is also reducing his profit potential. The risk/ reward ratio, hence, is an important consideration.

The risk of exercise in a spread position is the same as that in a short position. Certain investors may be able to anticipate exercise and execute a "rollover" transaction. However, should exercise occur, it would clearly mark the end of the spread position and thereby change the risk/reward ratio. Due to early assignments of the short side of the spread, what appears to be a limited risk spread may have more risk than initially perceived. An investor with a spread position in index options that is assigned an exercise is at risk for any adverse movement in the current level between the time the settlement value is determined on the date when the exercise notice is filed with OCC and the time when such investor sells or exercises the long leg of the spread. Other multiple-option strategies involving cash settled options, including combinations and straddles, present similar risk.

Important information: Examples within are indicative only, please call your local Morgan Stanley Sales representative for current levels.

By selling an option, the seller receives a premium from the option purchaser, and the purchase receives the right to exercise the option at the strike price. If the option purchaser elects to exercise the option, the option seller is obligated to deliver/purchase the underlying shares to/from the option buyer at the strike price. If the option seller does not own the underlying security while maintaining the short option position (naked), the option seller is exposed to unlimited market risk.

Spreading may entail substantial commissions, because it involves at least twice the number of contracts as a long or short position and because spreads are almost invariably closed out prior to expiration. Potential investors should carefully review tax treatment applicable to spread transactions prior to entering into any transactions.

Multi-legged strategies are only effective if all components of a suggested trade are implemented.

Investors in long option strategies are at risk of losing all of their option premiums. Investors in short option strategies are at risk of unlimited losses.

There are special risks associated with uncovered option writing which expose the investor to potentially significant loss. Therefore, this type of strategy may not be suitable for all customers approved for options transactions. The potential loss of uncovered call writing is unlimited. The writer of an uncovered call is in an extremely risky position, and may incur large losses if the value of the underlying instrument increases above the exercise price.

As with writing uncovered calls, the risk of writing uncovered put options is substantial. The writer of an uncovered put option bears a risk of loss if the value of the underlying instrument declines below the exercise price. Such loss could be substantial if there is a significant decline in the value of the underlying instrument.

Uncovered option writing is thus suitable only for the knowledgeable investor who understands the risks, has the financial capacity and willingness to incur potentially substantial losses, and has sufficient liquid assets to meet applicable margin requirements. In this regard, if the value of the underlying instrument moves against an uncovered writer's options position, the investor's broker may request significant additional margin payments. If an investor does not make such margin payments, the broker may liquidate stock or options positions in the investor's account, with little or no prior notice in accordance with the investor's margin agreement.

For combination writing, where the investor writes both a put and a call on the same underlying instrument, the potential risk is unlimited.

If a secondary market in options were to become unavailable, investors could not engage in closing transactions, and an option writer would remain obligated until expiration or assignment.

The writer of an American-style option is subject to being assigned an exercise at any time after he has written the option until the option expires. By contrast, the writer of a European-style option is subject to exercise assignment only during the exercise period.

Disclosure Section

Mortgage Backed Securities (MBS) and Collateralized Mortgage Obligations (CMO)

Principal is returned on a monthly basis over the life of the security. Principal prepayment can significantly affect the monthly income stream and the maturity of any type of MBS, including standard MBS, CMOs and Lottery Bonds. Yields and average lives are estimated based on prepayment assumptions and are subject to change based on actual prepayment of the mortgages in the underlying pools. The level of predictability of an MBS/CMO's average life, and its market price, depends on the type of MBS/CMO class purchased and interest rate movements. In general, as interest rates fall, prepayment speeds are likely to increase, thus shortening the MBS/CMO's average life and likely causing its market price to rise. Conversely, as interest rates rise, prepayment speeds are likely to decrease, thus lengthening average life and likely causing the MBS/CMO's market price to fall. Some MBS/CMOs may have "original issue discount" (OID). OID occurs if the MBS/CMO's original issue price is below its stated redemption price at maturity, and results in "imputed interest" that must be reported annually for tax purposes, resulting in a tax liability even though interest was not received. Investors are urged to consult their tax advisors for more information. Government agency backing applies only to the face value of the CMO and not to any premium paid.

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley & Co. LLC and/or Morgan Stanley C.T.V.M. S.A. and/or Morgan Stanley México, Casa de Bolsa, S.A. de C.V. and/or Morgan Stanley Canada Limited and/or Morgan Stanley & Co. International plc and/or Morgan Stanley Europe S.E. and/or RMB Morgan Stanley Proprietary Limited and/or Morgan Stanley MUFG Securities Co., Ltd. and/or Morgan Stanley Capital Group Japan Co., Ltd. and/or Morgan Stanley Asia Limited and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105), Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Stephan M Kessler; Mingyong Liu; Jakob Christian Lorenzen; Krystal Qian; Aristeidis Tentes; Vishwanath Tirupattur.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflict/policies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of December 31, 2025)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1519	41%	415	48%	27%	674	42%
Equal-weight/Hold	1583	43%	362	42%	23%	720	45%
Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	564	15%	87	10%	15%	216	13%
Total	3,670		865			1611	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley

processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 14-9169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i)

are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

The following authors are Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity securities: Vishwanath Tirupattur.

© 2026 Morgan Stanley